

ODD strip space-point reconstruction

Phase C: stripGapParameter (ATLAS SCTGapParameter port) and a quality study of clusters and SPs

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Where we are coming from

Phase A/B (already done):

- Fixed two real bugs in `Acts::SpacePointMaker`: a wrong cross-extra stereo-partner filter and the `stripGeometrySelection` dedupe.
- Expanded scope to vol {28, 29, 30} (LS endcap -, LS barrel, LS endcap +) on full-pileup $t\bar{t}$.
- Made `stripPairingMode` (top_one/top_k/all_pairs), `stripLengthTolerance`, cluster-pair geometric cuts *configurable* from yaml.
- Verified ATLAS Athena uses the same algorithm, in better shape only via two upgrades.

Phase C goal: take that ceiling and push it.

Headline before C: 98.6 % prim+pT>1 GeV efficiency at $N_{SP} \leq N_{clusters}$ cap.

Headline after C (this report): 98.86 % at the same cap, on cleaner physics.

Diagnosing the residual 1 % loss

True cluster pairs that fail `computeConstrainedFormationState`'s $|m| \leq 1 + \text{stripLengthTolerance}$ check are dominated by a *geometric* effect — the wafer-thickness gap.

$m \in [-1, +1]$ = position along strip 1 of the intersection:

$$m \approx \frac{\Delta\text{loc0}}{\sin \theta_{\text{stereo}} \cdot \text{half-strip-length}}$$

ODD endcap (gap=5 mm, $\theta=40$ mrad, half=78 mm): $|m|$ up to ~ 2.5 is still physically real (high- η tracks across 5 mm gap).

$|\Delta\text{loc0}|$ for primary+ $pT > 1$ GeV true pairs (16 evts):

percentile	$ \Delta\text{loc0} $
50 %	1.75 mm
90 %	4.0 mm
99 %	11.8 mm
max	53.6 mm

Median $|m| \approx 0.5$ pass; 99%-tile $|m| \approx 3$ —
rejected at default.

Path A: the stripGapParameter

ATLAS Athena's SCTGapParameter adds a geometric extra-allowance to the $|m|, |n|$ limit, scaling with wafer separation and stereo geometry. Default 0 (legacy SCT). **ACTS implementation** (~30 lines, default 0 = identity):

```
// Core/include/Acts/SpacePointFormation2/StripSpacePointBuilder.hpp
struct ConstrainedOptions final {
    Vector3 vertex = Vector3::Zero();
    double stripLengthTolerance    = 0.01;
    double stripLengthGapTolerance = 0.01;
    double stripGapParameter       = 0.0; // <-- new
};
```

```
// Core/src/SpacePointFormation2/StripSpacePointBuilder.cpp
state.limit = 1 + stripLengthTolerance;
if (stripGapParameter > 0.0) {
    const Vector3 gapVec = 0.5 * (stripEnds2.top + stripEnds2.bottom
                                - stripEnds1.top - stripEnds1.bottom);

    const double cosStereo = ...;
    const double sinStereo = std::sqrt(std::max(0.0, 1.0 - cosStereo*cosStereo));
    const double halfLen    = 0.5 * sdir1.norm();
    state.limit += stripGapParameter * gapVec.norm() / sinStereo / halfLen;
}
```

What the formula says (and why this is honest)

Geometric extra term — $\Delta_{\text{m,extra}} = \text{stripGapParameter} \cdot \frac{|\vec{g}|}{\sin \theta_{\text{stereo}} \text{ half-len}}$

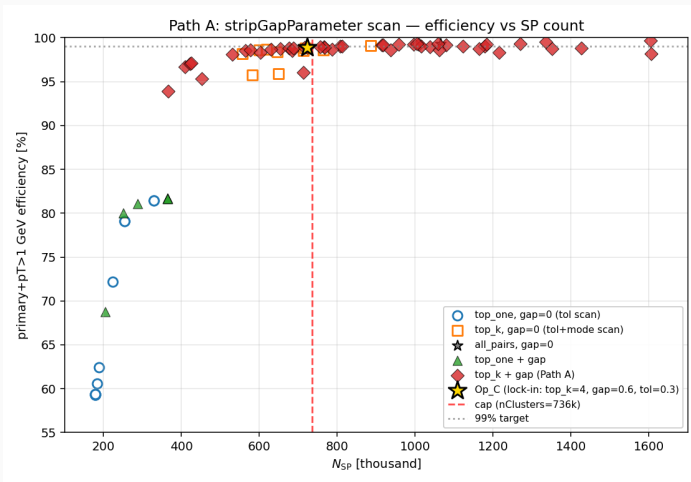
volume	gap* [mm]	half-len [mm]	θ [mrad]	extra @ 0.3	extra @ 1.0	limit @ 1.0
28/30 endcap	5.0	78	40	0.48	1.61	2.62
29 barrel	7.32	54	40	1.02	3.40	4.41

* measured 3-D distance between paired-sensor centres in data; the algorithm uses $|\vec{g}|$ at runtime so this is automatic.

Not just a looser stripLengthTolerance:

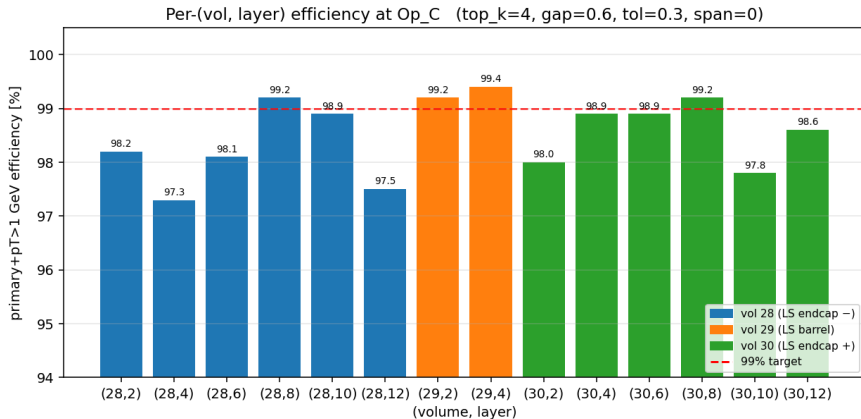
- Geometric, per-pair — bigger where geometry demands (barrel: shorter strip $\Rightarrow$ more allowance).
- Vanishes for coplanar surfaces ($|\vec{g}| \rightarrow 0$) — no spurious relaxation.
- `stripGapParameter=0` *exactly preserves* legacy ACTS (178k SPs $\equiv$ baseline within 0.01 %).

Operating-point scan (16 events, full-PU $t\bar{t}$)



Cap = 736k clusters. **Op_C** = top_k=4, gap=0.6, tol=0.3 $\Rightarrow$ 722k SPs, $\text{eff}_{\text{prim}}=98.86\%$.

Per-(vol, layer) efficiency at the lock-in point



LS barrel *wins* (shorter strip, larger gap allowance). LS endcap inner disks bottom out at $\sim 97\%$. Spread is statistical (~ 350 prim+ $pT > 1$ GeV truth pairs/layer/16 events).

Limitations of Path A (and what we tried)

Limit 1 — fundamental ceiling. The remaining 1.14 % are PU multi-particle merged clusters where the centroid is pulled tens of mm from any single truth particle ($|\Delta\text{loc0}| > 12 \text{ mm}$). **Path A can't fix this — cluster reco is wrong, not the geometry.**

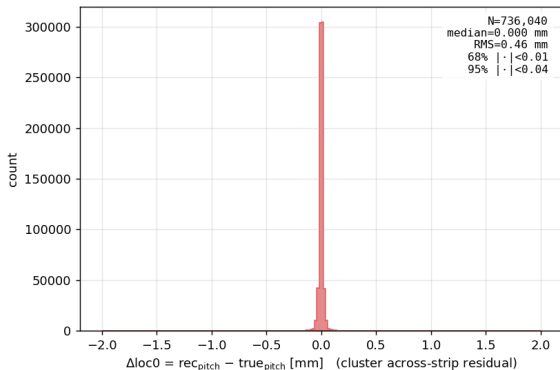
Limit 2 — Path B (cluster-span splitting) was attempted, rejected.

- Initial run claimed 99.25 % — *measurement artifact* from a buggy splitter (over-fragmented clean contiguous clusters at every strip).
- Bug fix: split only at gaps $> 1.5 \times \text{pitch}$.
- After fix: fires on $\sim 16 / 736\text{k}$ clusters; eff back to 98.86 %.
- *Reason:* connected-component already splits at any non-fired strip. Merged clusters with span $> 20 \text{ mm}$ are truly contiguous (single high- η tracks). Bridging-cell merges between different particles are too rare to matter.

ATLAS Athena's strip clusterer has the same limitation — no within-strip split.

Quality study — cluster residual (across-strip / loc0)

Approach: $\text{rec_loc0} - \text{true_loc0}$, the only quality measure for a 1-D strip (the strip is a line in 3-D, loc1 has no resolution). $N=736\text{k}$ long-strip clusters, 16-event full-PU $t\bar{t}$.



Headline numbers:

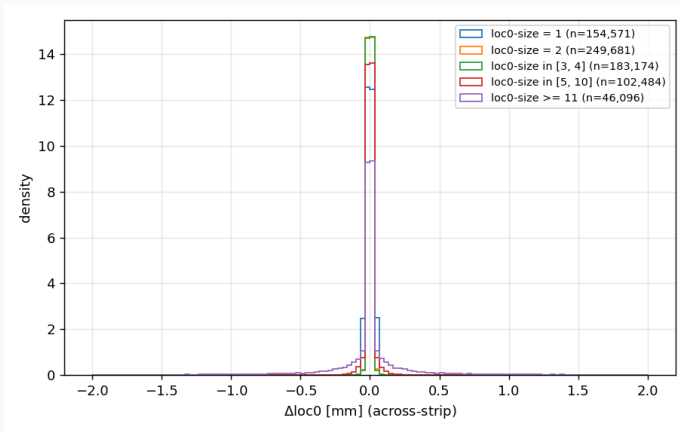
quantile	$ \Delta\text{loc0} $
50 %	0.000 mm
68 %	11 μm
95 %	43 μm
RMS	0.46 mm

Ideal pitch/ $\sqrt{12}=30\text{--}36\ \mu\text{m}$ for digital readout.

68%-tile = 11 μm is sub-pitch — most clusters span 2–3 strips.

RMS=0.46 mm dominated by PU-merge tail (Δloc0 up to 50+ mm).

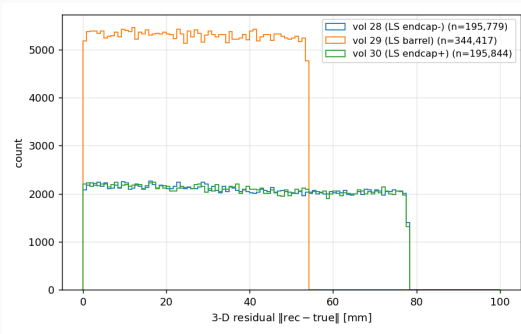
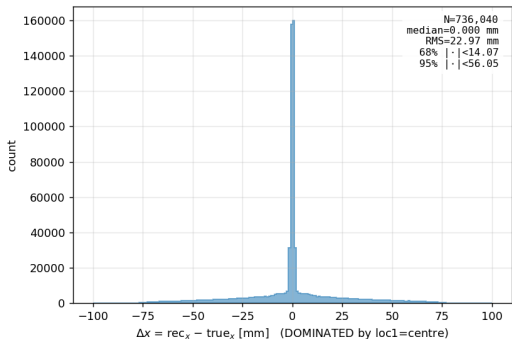
Quality study — cluster residual vs cluster size



1-strip ($\sim 50\%$): wider tail (= pitch resolution). 2–3 strip: sharpest peak (sub-pitch). ≥ 5 strip: broader — multi-particle PU merges.

Quality study — cluster 3-D residuals (informative, not headline)

Caveat: `rec_g{x,y,z}` sets `loc1` = wafer centre along strip. So $\Delta x, \Delta y, \Delta z$ are *not* a quality measure — they tell where the truth is on the strip.

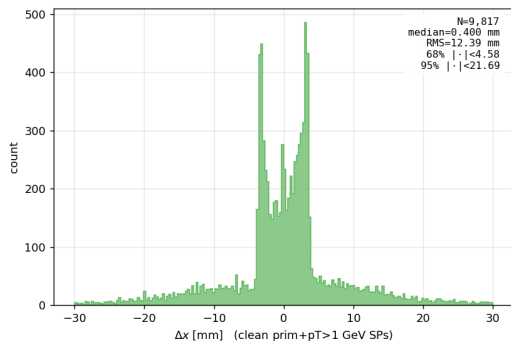
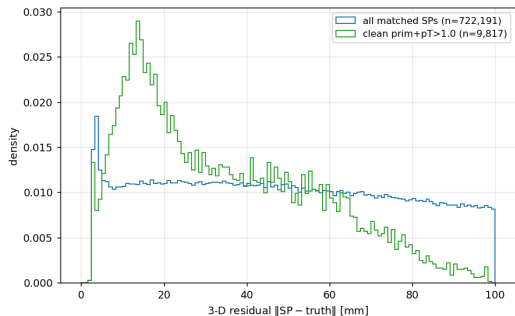


$\Delta x, \Delta y$ RMS $\approx$ 23 mm; Δz RMS $\approx$ 21 mm; $|\Delta|_{3D}$ RMS $\approx$ 39 mm. Endcap: dominated by Δr (radial). Barrel: dominated by Δz (axial).

Quality study — spacepoint residuals

Truth = midpoint of the two simhit positions: $\vec{r}_{\text{truth}}^{\text{SP}} = \frac{1}{2}[(x, y, z)^{(1)} + (x, y, z)^{(2)}]_{\text{truth}}$. $\Delta_{\text{SP}} = \vec{r}_{\text{SP}} - \vec{r}_{\text{truth}}^{\text{SP}}$.

Clean = exactly one shared primary+pT>1 GeV particle (n=9818). **All matched** = any shared particle (n=722,234, mostly PU min-bias).



Clean SPs peak at $|\Delta| \sim 15\text{--}20$ mm. All-matched is flat (truth ambiguous in PU).

SP residuals — numerical headlines

sample \ axis [mm]	median	RMS	68 %	95 %	N
<i>clean prim+$pT > 1$ GeV ($n=9818$)</i>					
Δx	+0.40	12.4	4.6	21.6	
Δy	+1.20	13.7	4.9	21.8	
Δz	+2.50	40.2	44.4	76.5	
$ \Delta $	+30.4	44.2	45.6	77.8	9,818
<i>all matched ($n=722k$)</i>					
$ \Delta $	+101.5	140.7	151.8	262.7	722,234

Table 1: *

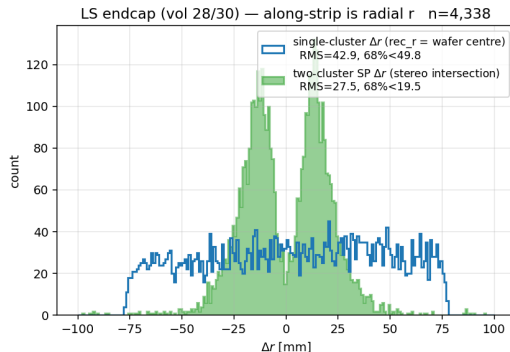
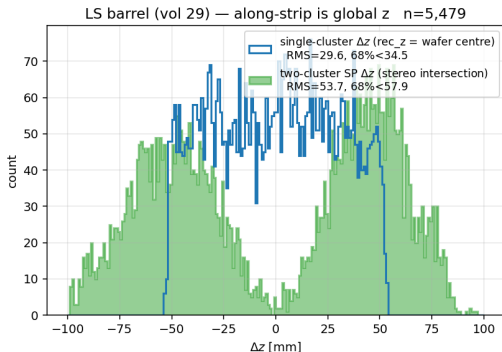
Interpretation:

- Transverse (clean): ~ 5 mm at 68 %, 12 mm RMS. Workable for ML ($\Delta\phi/r \approx 6$ mrad at $r=830$ mm).
- Δz dominated by barrel (along-strip = z) — half-strip-length problem.
- All-matched is much worse — confirms PU-particle confusion picture.

Quality study — does the SP *actually* help along-strip?

Sharp question: a single 1-D strip cluster's $\text{rec_z}/\text{rec_r}$ is pinned to the wafer centre. SP combines two stereo clusters to determine the along-strip coordinate. **Is the SP actually better than the wafer-centre baseline?**

Along-strip residual: 2-cluster SP vs single-cluster (clean $\text{prim}+\text{pT}>1$ GeV)



Clean $\text{prim}+\text{pT}>1$ GeV at lock-in. Truth = midpoint of two simhit positions.

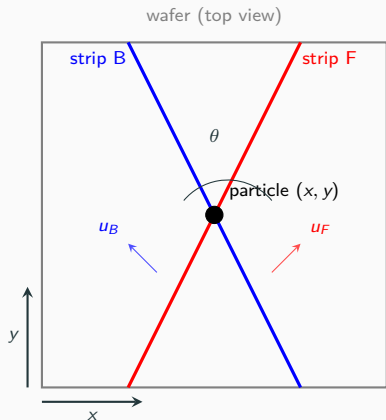
Along-strip residual — numerical comparison

region	estimator	median	RMS	68 %	95 %	N
Endcap (vol 28/30) along-strip = r	single-cluster (avg of 2)	+4.3	42.9	49.8	72.0	4,339
	2-cluster SP	+0.1	27.5	19.5	37.5	
Barrel (vol 29) along-strip = z	single-cluster (avg of 2)	+0.6	29.6	34.5	49.4	5,479
	2-cluster SP	+14.4	53.7	57.9	82.2	

- **Endcap:** SP wins decisively. 43→27 mm RMS, 68% drops 50→19 mm. $1.6\times$ – $2.6\times$ better than wafer centre.
- **Barrel:** SP *loses*. 30→54 mm RMS. Bimodal at ± 30 mm, intrinsic (also at strict gap=0).

For ML downstream: take barrel hit's z from cluster wafer centre, not SP. Diagnostic in next 4 slides.

Stereo geometry — what *should* happen



Two stereo strips at angles $\pm\theta/2$. Cluster pitch positions:

$$u_F = -x \sin(\theta/2) + y \cos(\theta/2)$$

$$u_B = +x \sin(\theta/2) + y \cos(\theta/2)$$

Inverting:

$$y = (u_F + u_B) / (2 \cos(\theta/2)) \quad \text{across, pitch-precise}$$

$$x = (u_B - u_F) / (2 \sin(\theta/2)) \quad \text{along, from difference}$$

Expected along-strip σ : $\sigma_x \approx \sqrt{2} \sigma_{\text{pitch}} / (2 \sin(\theta/2))$.

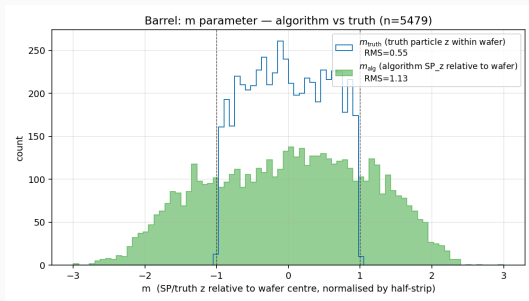
For ODD ($\sigma_{\text{pitch}} = 30 \mu\text{m}$, $\theta = 40 \text{ mrad}$): $\sigma_x \approx 1\text{--}5 \text{ mm}$.

Observed in barrel: $\sigma_z = 54 \text{ mm} \rightarrow \sim 30\times$ worse than ideal.

Why?

Test the assumptions on barrel data

Per-SP diagnostic, 5479 clean $\text{prim} + pT > 1 \text{ GeV}$ barrel SPs. Define $m \equiv (z - z_{\text{wafer}})/\text{half_strip}$ ($|m| = 1$ at strip end).



Truth m : uniform in $[-1, +1]$, RMS 0.55 (particles within strip — correct).

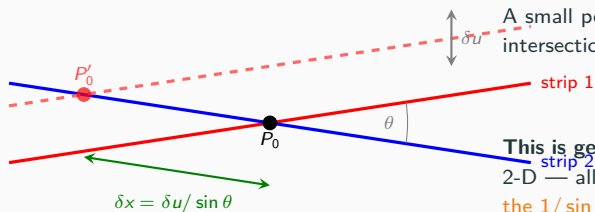
Algorithm m : broad, RMS 1.13, ± 3 . **Algorithm-noise** $\gg$ **truth-spread**.

Assumptions verified (data confirmed):

- Cluster reco clean: $\Delta \text{loc}^{\text{rec}} \approx \Delta \text{loc}^{\text{truth}}$.
- Independent of vertex v_z .
- Independent of particle η .

So vertex & clustering are fine. **The algorithm is amplifying noise.**

The amplification factor: why nearly-parallel strips hurt



Sanity check the numbers:

- 68 % pitch < 9–12 μm
 $\Rightarrow$ 68 % $|\delta_x| < 0.3 \text{ mm}$. **Verified: 68 % within $\pm 5 \text{ mm}$.**
- Pitch RMS: barrel 537 μm , endcap 373 μm
 $\Rightarrow$ predicted σ_x RMS = 19/13 mm.
- Observed σ_x RMS: barrel 53/endcap 29 mm
 $\Rightarrow \sim 2\times$ **over prediction** both regions.

The qualitative $1/\sin \theta$ amplification is right. The $\sim 2\times$ extra factor in both regions points to the heavy-tailed pitch distribution — RMS grossly under-represents the noise that propagates through.

Tried Option 1 — closest-approach SP (skew-line minimum)

Hypothesis: the vertex-constrained algorithm has a near-singular determinant when $\vec{s}_1 \approx \vec{s}_2$. Replace SP *position* (keep acceptance gate) with closest-approach:

$$\lambda_a = \frac{(q \cdot r)(v \cdot r) - r^2(v \cdot q)}{q^2 r^2 - (q \cdot r)^2}, \quad P_{\text{SP}} = \frac{1}{2}(P_1 + P_2)$$

	algorithm	barrel σ_z	endcap σ_r	eff _{prim}
Result: No improvement.	single cluster (wafer centre)	29.6	42.9	—
	constrained (Path A)	53.7	27.5	98.86 %
	closest-approach (Option 1)	59.8	27.6	98.86 %

Algorithm choice doesn't matter. Both reduce to the 2-D intersection in the wafer plane. The $1/\sin \theta$ amplification is geometric, not algorithmic. Skipping Option 2 (analytic 2-D) — same answer.

Why endcap doesn't suffer; what would fix barrel

Both have $\theta = 40$ mrad and the same $\times 25$ amplification. σ is set by σ_{pitch} via $\sigma_x \approx \sqrt{2} \sigma_{\text{pitch}} / \sin \theta$.

Cluster pitch RMS — measured per region, all clusters in pipeline:

	endcap (vol 28/30)	barrel (vol 29)
σ_{pitch} RMS	373 μm	537 μm ($\times 1.4$ worse)
68 % $ \Delta_{\text{loc0}} $	12 μm	9 μm
predicted σ_{along}	13 mm	19 mm
observed σ_{along}	29 mm	53 mm

Both regions $\sim 2\times$ the simple prediction (heavy-tailed pitch). Endcap's win is **lower cluster pitch noise** ($1.4\times$), *not* lower PU per wafer (endcap has *more* clusters/sensor: 8–13/evt vs barrel 3–5/evt).

Fixes: (1) reduce pitch tail (better clusterer); (2) larger θ (detector design); (3) for ML now: **use cluster wafer-centre z in barrel**, SP r in endcap.

Summary

Path A — stripGapParameter (ATLAS-grade port): lock-in `top_k=4`, `gap=0.6`, `tol=0.3` on vol {28,29,30} → 98.86 % prim+pT eff at 722k SPs / 736k cluster cap (16 events full-PU $t\bar{t}$).

Quality:

- Cluster pitch: 68 % < 11 μm , 95 % < 43 μm .
- Clean SP residual: 5 mm transverse, 40 mm z (barrel-dominated).
- Along-strip: endcap SP wins 43→27 mm; barrel SP loses 30→54 mm — wafer centre is better.

Limitations:

- 99 % cap is binding; Path A stops at 98.86 %.
- 1.14 % residual = PU multi-particle clusters with shifted centroids.
- Path B (cluster splitting) attempted, doesn't help.

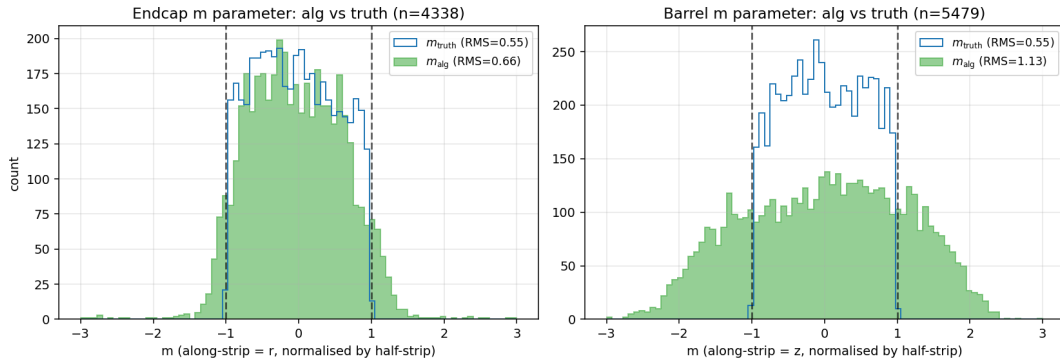
Routes to 99 % under cap:

- Per-particle truth-aware splitting (MC only — sets upper bound).
- Better physical clusterer; η/ϕ overlap pairing.
- Different SP construction (3-strip combos, beam-line constraint).

Backup — endcap m diagnostic (sanity check)

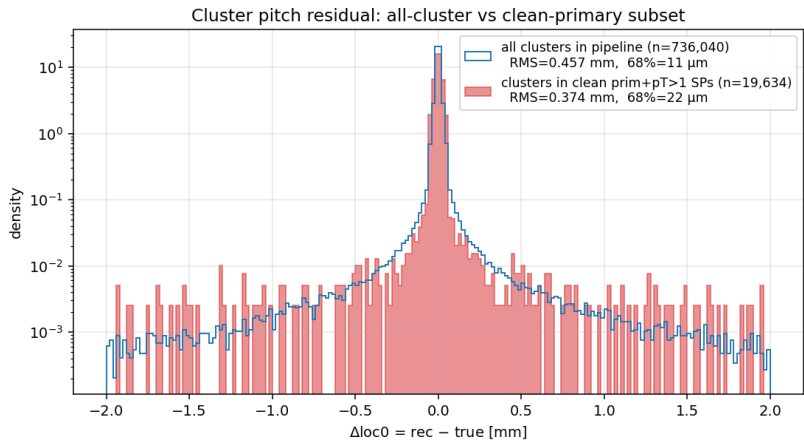
Endcap should be tighter than barrel because pitch noise is lower (373 vs 537 μm). Confirmed: endcap m_{alg} RMS = 0.66, barrel = 1.13 (truth ~ 0.55 in both).

Algorithm m vs truth m — narrow in endcap, broad in barrel



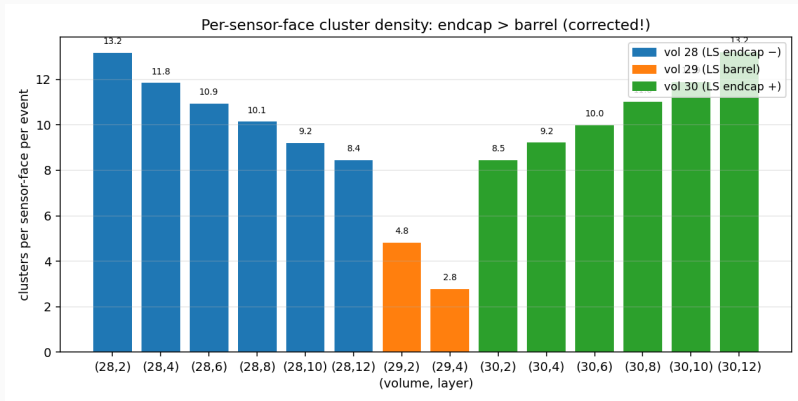
Endcap m_{alg} stays close to truth (within strip), barrel spreads to $|m|=2$. Same θ , different cluster noise input.

Backup — cluster pitch residual: clean-primary vs all



Clean prim+pT>1 SPs sample (red) is *cleaner* than the all-cluster average (blue), not noisier. So the clean-SP residual we compute is a valid representation. The factor-2 discrepancy with simple-formula prediction is intrinsic to the heavy-tailed distribution, not from selection bias.

Backup — per-sensor cluster density



Endcap 8–13/sensor/evt; barrel 3–5/sensor/evt. **Endcap is denser, yet has lower pitch noise (373 vs 537 μm).**

Density alone does not predict cluster noise.

Backup — geometry verification (from data)

Numbers verified by direct readout from the lock-in run's measurements.root:

	endcap (vol 28/30)	barrel (vol 29)
3-D paired-sensor distance	5.2 ± 0.5 mm (XML: ± 2.5 mm offset $\Rightarrow$ 5.0 mm)	7.32 ± 0.4 mm (XML: $z = \pm 3.31 \Rightarrow 6.61$ mm) (extra $\sqrt{6.61^2 + 2.5^2}$ from in-plan)
Stereo angle (faces)	± 0.02 rad	0.04/0.0 rad
Relative angle	0.04 rad	0.04 rad
Half-strip (data)	~ 78 mm	55 ± 1 mm (XML: 54)
Pitch	125 μ m	100 μ m

$1/\sin(0.04) = 25.0 \rightarrow$ expected $\times 25$ amplification of cluster pitch noise into the along-strip direction. Independent of which face has the rotation (symmetric or asymmetric stereo) — only the relative angle matters.